Spectre Defense on RISC-V — From gem5 Simulation to RTL Implementation

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Abstract

This is the midpoint report, detailing our approach, preliminary results, and remaining steps.

1 Introduction

The field of RISC-V security is still a open topic: even with gem5 simulation validating security across different defenses, the questions of area and performance are still relevant to understand the potential impacts of these security features. By exposing commercial RISC-Vs are still prone to Spectre, there is a growing demand to systematically classify software and hardware defenses for RISC-V [3]. Speculative taint tracking is one such defense done on existing x86 and ARM architecture. Could we implement this in RISC-V hardware?

The major questions arises: what sort of Spectre defenses could we implement, how do we evaluate these tradeoffs, and how close does the gem5 simulations match the metrics on a RISC-V implmentation?

2 Background

Spectre attacks exploit the discrepancy between architectural correctness and micro-architectural state, allowing transiently executed instructions to leak sensitive information through cache side channels [6]. Existing defenses have been evaluated extensively using architectural simulators such as gem5 [7], but simulation abstractions may not fully capture the pipeline timing, cache replacement behavior, and hardware resource contention of synthesizable RISC-V processors. In evaluating different software defenses in SpectreShield [5], gem5 was used as the main simulation tool to observe speculation behavior on micro-architectural elements. Here, they explicitly list out the various metrics used to evaluate Spectre in cache contention, branch prediction accuracy, DRAM access & power variability, and secret key recovery rates

Prior research has implemented speculative-execution defenses in both gem5 and RISC-V for caches (SpecLFB) [2] and taint-tracking (Secure Speculative Execution) [8]. In addition, SecureBOOM semi-automates integrating security features into the microarchitecture without having the designer consider every potential security gap; this is done through formal analysis on the RTL design, which was implemented on the Berkeley Out-of-Order Machine (BOOM). Important questions remain regarding the consistency of their security guarantees and performance overhead across these abstraction levels [4]. Indeed, uSTT mentions how SecureBOOM's infrastructure negatively impacts area overhead [1].

This project proposes an implementation of an existing speculation-aware cache defense on an open-source RISC-V processor and evaluates it against a corresponding gem5 reference model. Specifically, we take a look at uSTT and implement it onto a RISC-V. By comparing security outcomes, cycle-level behavior, performance degradation, and hardware costs, we aim to determine where gem5 accurately represents RTL behavior, characterizing the trade-offs between security and processor efficiency.

3 Threat Model

What is the current architecture? (SonicBOOM on Xilinix FPGAs) How do we set up the board for the attack? What sort of attacks are we trying to implements? What sort of defense are we trying to implement? What assumptions do we make on the attacker side? Describe your threat model.

4 Our Approach

Describe your approach to the problem.

5 Evaluation & Preliminary Results

Explain your evaluation setup and preliminary results.

6 Remaining Steps

Remaining Steps

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